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Tangible Media & Physical Computing
CART 360, Section AA

InFringement
Prototype: Everywhere = Nowhere = Now

Work presented to
Elio Bidinost

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## 1. Project Description

*InFringement* is a small-scale installation and interactive experience that is designed to explore the themes of human interaction, self-awareness, and gender performativity. Two strangers are chosen to participate in the experience at the same time. They will each wear a colourful, fun, feminine wig, which will be equipped with GSR sensors and stepper motors. The participants will then come face-to-face and are allowed to speak to each other, while observers in the room stay quiet. The wigs will respond to each wearer's stress levels, moving the bangs up or down, creating a dynamic visual representation of the participants' emotional states that become immediately visible to both participants and observers.

The mechanical system we will create will transform invisible physiological signals into obvious motion. GSR sensors measure increased skin conductivity caused by sweat gland activity, which is controlled by the sympathetic nervous system. When someone is stressed, their sweat gland secretions increase, and with that increase moisture, the electrical resistance on skin reduces. The stepper motors translate this data into movement of the wig bangs. Each wig reacts independently, which creates a personal feedback loop for the wearer, while observers watch the interaction without directly intervening. The movement of covering and revealing the wearer's eyes emphasizes vulnerability, attention, and emotional presence in the interaction.

By allowing participants to speak while observers stay silently present, the experience highlights the complex connections between communication, self-expression, and social tensions. The two participants are set up to navigate their own emotional responses while handling the performative demands of the situation, reflecting on how they present themselves and respond to subtle social cues. Additionally, the presence of mute observers intensifies this dynamic through the creation of a heightened sense of being watched, another factor that can increase stress and self-consciousness. Simultaneously, wearing a wig that moves unpredictably transforms this pressure into an opportunity for playful experimentation with identity and gender expression, particularly for participants who might not typically engage with traditionally feminine aesthetics.

The interaction strategy relies on the human body as the link between sensor input and mechanical output, which makes the system inherently unpredictable. Each session unfolds differently because participants' emotional responses, speech, and awareness of being observed all influence how the wigs move in real time. The slightly absurd movement of the wigs is intentional, encouraging participants to notice their own reactions and engage with the situation in a playful, reflective way. Over multiple sessions, both participants and observers may start to pick up on patterns in behavior and stress responses, highlighting how subtle feedback loops emerge when social dynamics are combined with a responsive, mechanical system.


## 2. Four Research Questions

The project will be presented as a small installation / interactive experience. Two strangers will be selected to participate in the experience by wearing the wigs. We will place the wigs on their heads and set up the GSR sensors. Then, they will face each other, with the freedom to speak to each other, while observers in the room remain silent. Their wigs respond to their stress levels moving the bangs up and down, reacting to the real-time data. *InFringement* is designed for people interested in exploring human interaction, social behaviour, and gender performance in a visual, playful, and reflective way.

The interaction focuses on self-awareness and reflection. By wearing the wig, each participant experiences their own stress in a tangible and physical way, with the bangs moving up or down depending on their GSR readings. This creates a situation where the participants have to navigate their own emotional states and the social dynamics of interacting with a stranger, while their emotional responses are visually displayed. It is meant to intentionally make people notice typically subtle emotions and how they show up in a shared, playful moment.

Our artifact challenges users in a way that could be uncomfortable to some. It puts people in a vulnerable position: interacting with a complete stranger, while wearing a wig that behaves and feels unusual. However, it also gives participants a way to play with identity and gender expression, especially since the wigs are colourful and feminine-looking. They might feel exposed at first, but that discomfort is the point because it pushes them to  reflect on how they perform their own gender and respond to social pressure, internally and externally.

The interaction is designed as a wearable, tangible feedback system. The GSR sensor measures each participant's stress, and the stepper motor moves the wig bangs up or down in response. This movement turns mostly invisible or inconspicuous emotions into visible motion. Because each wig reacts only to its wearer, participants experience a personal feedback loop and the social setup of facing a stranger adds tension and awareness. The bangs serve as an expressive interface, by covering or revealing the eyes which manifest vulnerability, connection, and exposure. This creates a multidimensional, embodied experience that encourages participants to reflect on their emotions, presence, and gender performativity.

## 3. Non-Technical Evaluation of Sensors

Physical input from two galvanic skin response (GSR) sensors connected to a pair of electrodes drive the device's behaviour unaccompanied throughout the span of the experience. A GSR sensor measures how electrically conductive skin is in contact with its electrodes. Effectively, this is a measure of how much sweat is being produced. This acts as a basic entry point into the user's emotional state, allowing us to generalize biological states to use in a myriad of ways.

## 4. Interaction Design Strategy

Although there is only one input method, *InFringement*'s interaction design is quite complex due to the human body that stands between the sensor and the mechanics. The action that begins the interaction between the device and its wearer is the GSR sensors' contact with the wearer's skin. In a vacuum, one may assume that the data received by these sensors would be independent from sources outside of the connected individual. However, by introducing an element of emotional disruption, we also have an indirect signal that has the opportunity to be dependent on itself.

Incorporating the human nervous system into the system adds a level of uncertainty that will change the experience each time it runs. Since this project relies on the psychology of each individual user, it is impossible to predict with complete certainty what environmental factors will apply to the mechanical output. With the complexity of the human brain in combination with often erratic human behaviour, the interaction strategy is most accurately described as a conversing system.

Its aesthetic absurdity is used as an intentional feature of the device. The intention is to demonstrate and physically manifest uncomfortability while encouraging it. By placing two wig wearers in front of each other, we expect each instance to showcase behaviour of a learning system. After a time, it is likely that the brain will no longer respond to the element of surprise in the same way as it initially does. Observing this mildly Pavlovian, Piaget-esque behaviour is the core of this entire project.

## 5. Similar Projects Research

To have a better understanding on how artists have used biofeedback and interactive systems in their work, we researched several related projects.

The first one we found interesting is *Biofeedback Artwork* (2011) by Amy Karle. It is a performance and video art piece in which the artist connects her body to technology to generate visual and auditory output in real time. While meditating, Amy Karle feeds her biofeedback signals into a Sandin Image Processor, which is an analog computer that was used for experimental video synthesis. The system translates the data into shifting images and sound, making internal bodily states visible through the video. The work explores the relationship between the human body, consciousness, and technology, by turning subtle biological processes into an expressive audiovisual experience.

The second artwork we found intriguing is *My Heart Laid Bare* (2011) by Anna Dumitriu, *My Heart Laid Bare* is a participatory performance installation that used a GSR sensor to visualize the artist's emotional state during conversations with gallery visitors. While sitting across from participants, the artist's physiological indicators were monitored and used to trigger video projections that "revealed her inner emotional state". As participants spoke with her, their words and behaviour could influence her emotional responses, creating a feedback loop between conversation, physiological data, and the visual output. The work explores themes of emotional transparency, surveillance, and the ethical implications of technologies that claim to reveal or interpret human feelings.

The final project we chose is *Third Hand* (1980) by Stelarc. *Third Hand* is a performance artwork in which the artist implements a robotic third hand to his body as a form of technological enhancement. The prosthetic hand is controlled by electromyography (EMG) signals from the artist's abdominal and leg muscles, which allows it to move independently from his natural hands. Through the performances, Stelarc explores the relationship between the human body and technology, and tries to imply that technological prosthetics can extend or enhance the body rather than simply replace missing parts.

## 6. Project Impact and Comparison

Compared to these projects, our work focuses less on visualizing internal states through complex audiovisual systems or technological body augmentation, and more on the social dynamics that emerge when biofeedback becomes visible during a real interaction between two people. Compared to Amy Karle's work which translates biofeedback into abstract visual media, and Anna Dumitriu's performance which reveals the artist's emotional data during conversation, our projects situates both participants under the same circumstances simultaneously, allowing their stress levels to unfold in a shared social space. Similarly, unlike Stelarc's *Third Hand*, which explores technological extension of the human body, *InFringement* uses relatively simple mechanics to highlight emotional and social responses rather than physical enhancement. Our project's goal is to create a playful and reflective situation where participants are forced to become aware of how stress, communication, and self-awareness interact in real time through the combination of biofeedback, a slightly absurd wearable piece of technology, and a silent watching audience.

## 7. Prototype Implementation and Team Responsibilities

Project Repository: https://github.com/mvlamis/wig

The github repository will contain the micropython code wiring references, and documentation related to the prototype's development.

Team members:

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Individual Contributions

Mars

- Project description
- Four research questions
- Similar projects research
- Project impact and comparison section
- Prototype testing

Michael

- Non-technical evaluation of sensors
- Interaction design strategy
- Project storyboard / illustrations
- Prototype testing
- Sensor programming / artifact building

**Bibliography**


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